Lab Assignment 7

Title :

Design and implement a hash table of fixed size. Use the division method for the hash function and resolve collisions using linear probing.

Program :

```
#include<iostream>
using namespace std;
#define SIZE 10

int main()
{
  int hashTable[SIZE];
  int key,index;

  for(int i=0;i<SIZE;i++)
   hashTable[i] =-1;

   cout<<"Enter 5 key:"<< endl;
    for(int i=0;i<5;i++)
    {
    cin>> key;

    index=key%SIZE;

    // linear probing for collision
    while(hashTable[index] !=-1)
    {
    index=(index+1)%SIZE;
    }
     hashTable[index]=key;
     }

     //display hash table
     cout<<"\nhash table:\n";
     for(int i=0;i<SIZE;i++)
     {
      cout<<i<<":";

      if(hashTable[i]==-1)
         cout<<"Empty";
      else
      cout<<hashTable[i];
      cout<< endl;
      }

      //search for key
      cout<<"Enter key to search:";
      cin>>key;

       index=key%SIZE;
       int start=index;
       bool found= false;
```

```
while(hashTable[index]!=-1)
{
if(hashTable[index]==key)\
{
 found=true;
 break;
 }
 index=(index+1) % SIZE;
  if(index==start)
  break;
  }
 if(found)
   cout<<"key found at index:"<<
   index<<endl;
   else
   cout<<"key not found:"<<endl;
   return 0;
   }
```

Output:

Output

```
Enter 5 key:
5
8
9
15
4

hash table:
0:Empty
1:Empty
2:Empty
3:Empty
4:4
5:5
6:15
7:Empty
8:8
9:9
Enter key to search:15
key found at index:6
```